VIBROACOUSTICS - ASSIGNMENT 1

Transverse Vibrations of a Tensioned String

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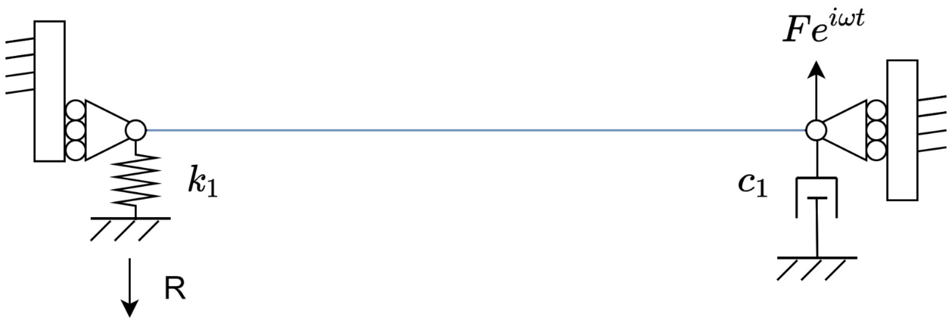
1 Introduction

This report addresses the first step of a three-part project aimed at analytically modelling the sound radiation from a mechanical system inspired by musical instruments. The full system consists of two coupled continuous elements: a tensioned string and a thin plate, connected through a localized elastic and damping element that represents the mechanical behaviour of a bridge.

The present assignment focuses exclusively on the transverse vibration of the tensioned string, treated as a standalone subsystem. The goal is to characterize its dynamic response under harmonic excitation. In particular, the following objectives are pursued:

- To determine the natural frequencies and mode shapes of the string using the standing wave solution, within the frequency range [0, 2000 Hz];
- To compute the driving-point receptance (at the free end of the string) and the transfer receptance (at the connected end);
- To evaluate the reaction force transmitted from the string to the plate, which will be used as the input for the second step of the project.



System Parameters

The physical parameters used for the simulation are:

- Tension: $T = 877$ N
- Density: $\rho = 7850$ kg/m^3
- Diameter: $d = 1$ mm $\Rightarrow$ Cross-sectional area: $A = \pi(d/2)^2$
- Length: $L = 1.5$ m
- Spring stiffness: $k_1 = 1.12 \times 10^8$ N/m
- Damping coefficient: $c_1 = 0.3$ Ns/m (used later)
- Excitation force: $F = 1$ N

2 Natural Frequencies and Mode Shapes

The transverse vibration of the string is governed by the classical one-dimensional wave equation:

$$\frac{\partial^2 w(x,t)}{\partial x^2} = \frac{1}{c^2}\frac{\partial^2 w(x,t)}{\partial t^2},$$

where $w(x,t)$ is the transverse displacement, and $c = \sqrt{T/(\rho A)}$ is the wave speed along the string. Here, T is the tension, ρ the material density, and A the cross-sectional area of the string.

We assume a standing wave solution of the form:

$$w(x,t) = \Phi(x) \cdot G(t),$$

which leads, via separation of variables, to the spatial equation:

$$\Phi''(x) + k^2\Phi(x) = 0, \quad \text{with } k = \frac{\omega}{c}.$$

The general solution is:

$$\Phi(x) = A\cos(kx) + B\sin(kx).$$

2.1 Boundary Conditions and Frequency Estimation

The system features a spring support at the left end ($x = 0$) and a damped support at $x = L$. The boundary conditions are:

- $T\frac{\partial w}{\partial x}(0,t) - k_1 w(0,t) = 0$ (elastic constraint)
- $T\frac{\partial w}{\partial x}(L,t) = 0$ (fixed end)

By applying these conditions to the spatial solution $\Phi(x)$, we form the matrix equation:

$$H(\omega) = \begin{bmatrix} -k_1 & T\gamma \\ -T\gamma\sin(\gamma L) & T\gamma\cos(\gamma L) \end{bmatrix}, \quad \text{with } \gamma = \frac{\omega}{c}.$$

Non-trivial solutions exist only when $\det H(\omega) = 0$, which defines the natural frequencies of the system. Since a closed-form expression is not available, we evaluate $|\det H(\omega)|$ numerically over the range [1, 2000] Hz and identify its local minima.

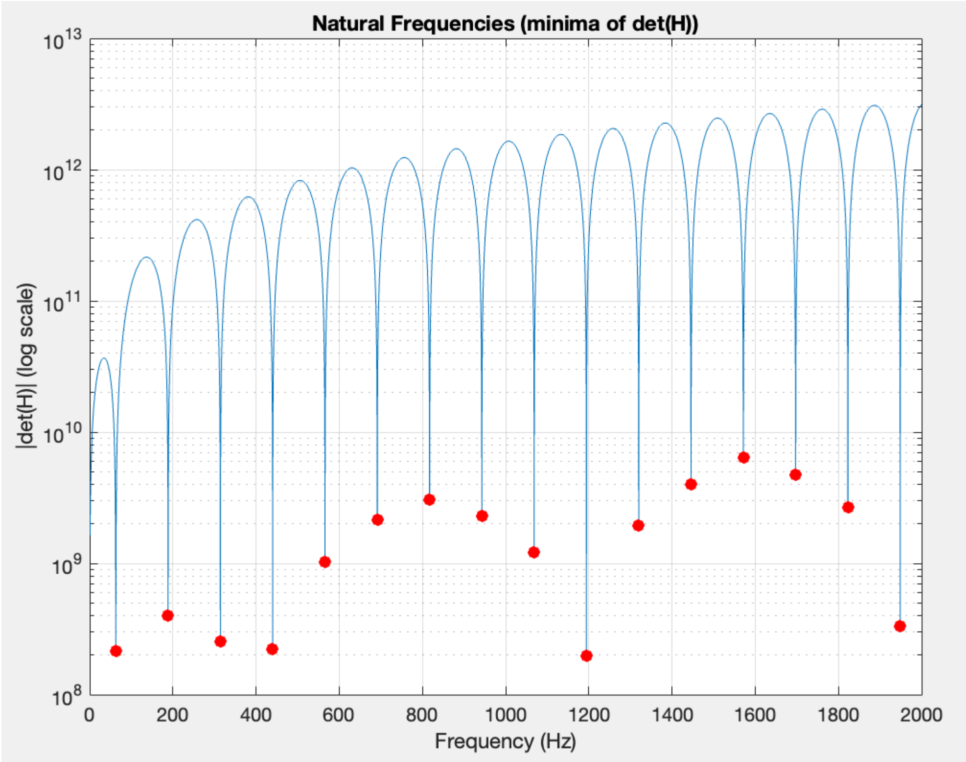


Figure 1: Logarithmic plot of $|\det H(\omega)|$ over the frequency range [0, 2000 Hz]. Each sharp minimum (red marker) corresponds to a natural frequency of the system. These values identify the resonance conditions under the applied boundary constraints.

2.2 Computation and Visualization of Mode Shapes

For each identified natural frequency f_i, the corresponding wavenumber is computed as:

$$k_i = \frac{2\pi f_i}{c}.$$

The general form of the i-th mode shape is

$$\Phi_i(x) \;=\; A_i \cos(k_i x) + B_i \sin(k_i x),$$

where the constants A_i and B_i satisfy the elastic boundary condition at $x = 0$:

$$T\,\Phi_i'(0) = k_1\,\Phi_i(0) \quad\Longrightarrow\quad B_i = \frac{k_1}{T\,k_i}\,A_i.$$

Note that A_i is free up to an arbitrary multiplicative constant—eigenvectors are defined only up to scale—so we fix A_i by our chosen normalization.

Each mode shape is then normalized with respect to its maximum absolute value,

$$\Phi_i(x) \;\leftarrow\; \frac{\Phi_i(x)}{\max_{x\in[0,L]}\big|\Phi_i(x)\big|},$$

and evaluated along the domain $x \in [0, L]$. The following figure shows the first five normalized shapes.

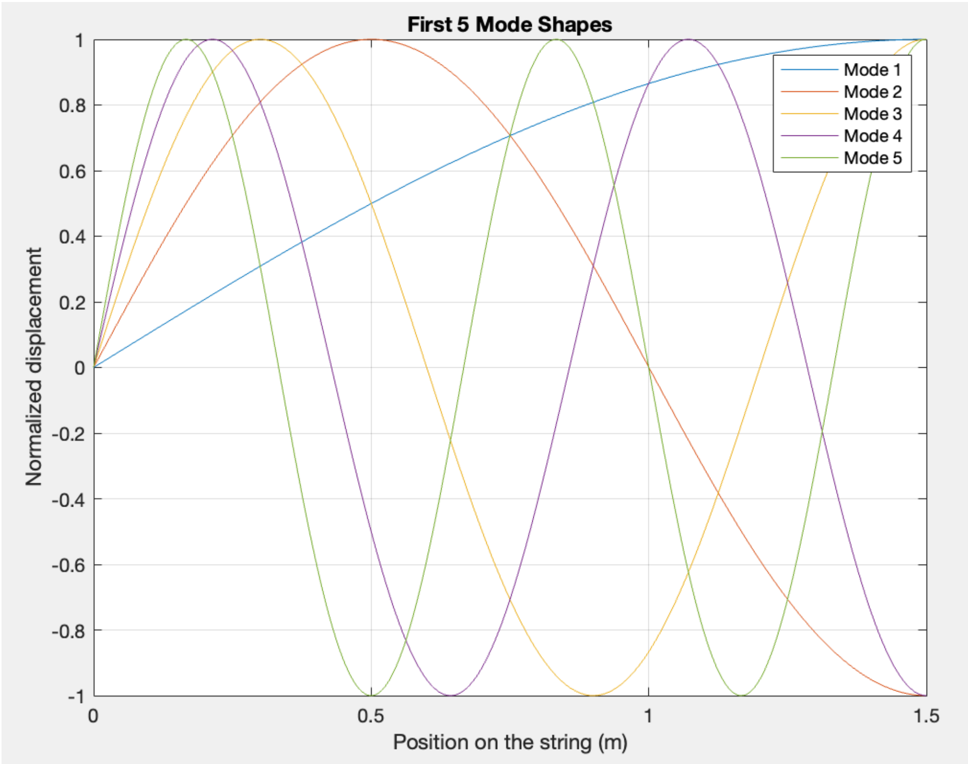


Figure 2: First five normalized mode shapes of the string with an elastic boundary at $x = 0$. Each mode exhibits slight asymmetry and nonzero displacement at $x = 0$.

- The mode shapes exhibit a nonzero displacement at $x = 0$ ($\Phi_i(0) \neq 0$), since the elastic boundary there allows movement. A rigid clamp instead would enforce $\Phi_i(0) = 0$.
- The number of **nodes** (points of zero displacement) increases with the mode index, as predicted by classical vibration theory. The first mode has no interior node, the second has one, the third has two, and so on.
- The relative **amplitudes** have no physical meaning in this plot since all shapes are normalized. However, they still illustrate the **distribution of energy** along the string and highlight the regions of maximal motion.
- The **curvature increases** with mode number, consistent with the higher spatial frequency content and greater strain energy of higher modes.

The resulting mode shapes exhibit increasing curvature and number of nodal points with increasing frequency, as expected. The first mode shows a single lobe without internal nodes, while the fifth mode already displays multiple inflection points.

3 Receptance Functions

In this section, we analyze the dynamic behavior of the string by computing its frequency-domain receptance functions under harmonic excitation.

For each angular frequency $\omega = 2\pi f$, we solve the frequency-domain boundary value problem using the following linear system:

$$H(\omega) = \begin{bmatrix} -k_1 & T\gamma \\ -T\gamma\sin(\gamma L) + j\omega c_1\cos(\gamma L) & T\gamma\cos(\gamma L) + j\omega c_1\sin(\gamma L) \end{bmatrix}, \quad \text{with } \gamma = \frac{\omega}{c}$$

The unknown vector $\begin{bmatrix} A \\ B \end{bmatrix}$ defines the coefficients of the standing wave solution $\Phi(x) = A\cos(\gamma x) + B\sin(\gamma x)$, derived from the general solution to the wave equation. For each frequency, the system is solved as:

$$H(\omega) \cdot \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} 0 \\ F \end{bmatrix}$$

Once the modal coefficients are known, we compute the receptance functions as follows:

- **Transfer receptance** (response at $x = 0$ due to force at $x = L$):

$$G_{\text{transfer}}(\omega) = \frac{w(0)}{F} = \frac{A}{F}$$

- **Driving-point receptance** (response at $x = L$, point of excitation):

$$G_{\text{drive}}(\omega) = \frac{w(L)}{F} = \frac{A\cos(\gamma L) + B\sin(\gamma L)}{F}$$

3.1 Driving-Point Receptance at $x = L$

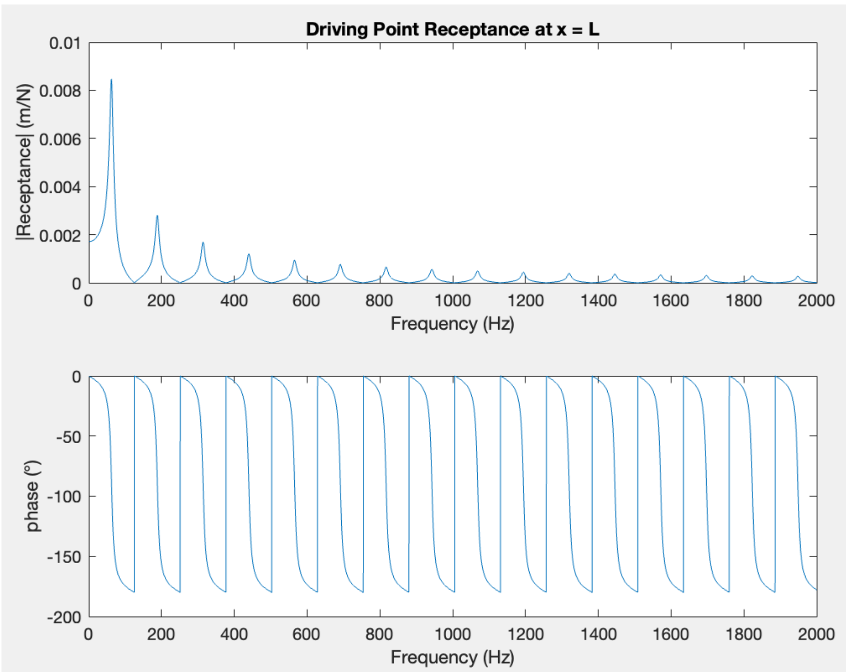


Figure 3: Magnitude and phase of the driving-point receptance at $x = L$.

As shown in the figure, the magnitude of the driving-point receptance exhibits pronounced peaks at the natural frequencies of the string. These correspond to resonance conditions, where the displacement response is maximized at the point of excitation. The first resonance is dominant, while the amplitude of the peaks gradually decreases with frequency.

The phase undergoes a downward shift of approximately 180° at each resonance, which is a classical behavior of second-order resonant systems. These transitions confirm the modal structure of the response.

Effect of Damping

The viscous damper c_1 located at $x = L$ introduces a damping force of the form:

$$F_{\text{damping}} = c_1 \cdot \frac{\partial w}{\partial t} = c_1 \cdot v$$

Under harmonic excitation, the velocity v is proportional to the displacement amplitude and the angular frequency ω, i.e., $v \propto j\omega \cdot w$. Consequently, the damping force becomes frequency-dependent:

$$F_{\text{damping}} \propto c_1 \cdot \omega \cdot \Phi(x)$$

This implies that at higher frequencies, the damping force increases proportionally, and its energy dissipation effect becomes more significant. This explains the observed attenuation in the receptance magnitude for higher modes: although resonance still occurs, the amplitude is progressively reduced due to the increased damping force.

This frequency-dependent damping behavior is a key characteristic of continuous vibrating systems with viscous boundary conditions. It is especially visible in the driving-point receptance, where the local velocity is high and the damping acts directly on the motion. Despite this, the natural frequencies remain essentially unchanged, as damping in this case is not strong enough to shift the modal frequencies appreciably.

3.2 Transfer Receptance at $x = 0$

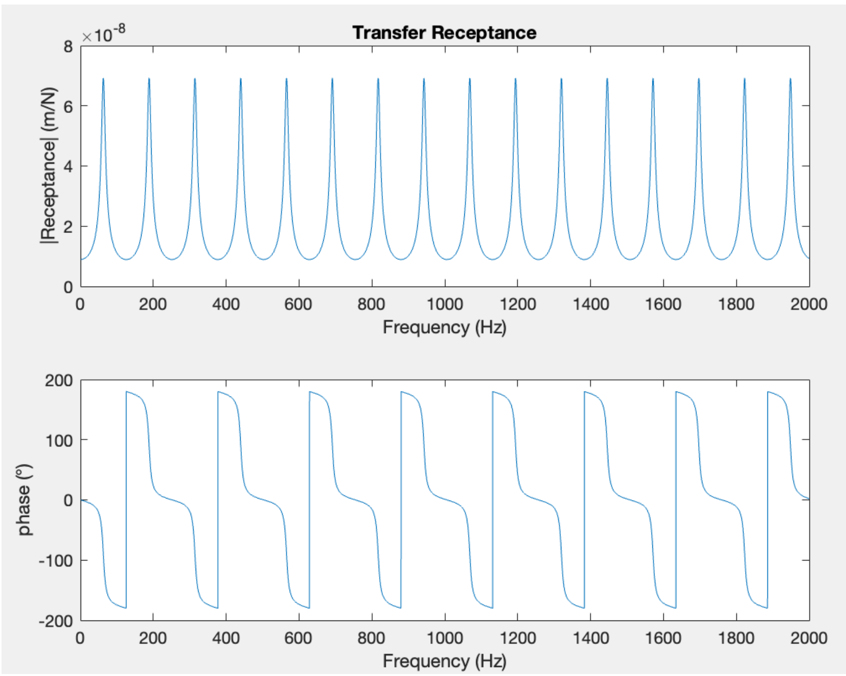


Figure 4: Magnitude and phase of the transfer receptance at $x = 0$.

In contrast, the transfer receptance measured at $x = 0$ exhibits significantly lower amplitude—several orders of magnitude smaller than the driving-point response. This is physically justified, as the excitation is applied at the opposite end of the string.

Nonetheless, the transfer receptance still shows resonant peaks aligned with the natural frequencies, indicating that the modal structure of the system remains fully coupled across the string.

4 Reaction Force

Finally, we evaluated the reaction force transmitted from the string to the adjacent plate. This force acts at $x = 0$, where the string is connected to a linear spring of stiffness k_1.

From the solution of the system at each frequency, the displacement at $x = 0$ is given by $w(0) = A$, where A is the coefficient obtained from the system:

$$H(\omega) \cdot \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} 0 \\ F \end{bmatrix}$$

Thus, the frequency-dependent reaction force is computed as:

$$R(\omega) = k_1 \cdot w(0) = k_1 \cdot A \tag{1}$$

Frequency Response of the Reaction Force

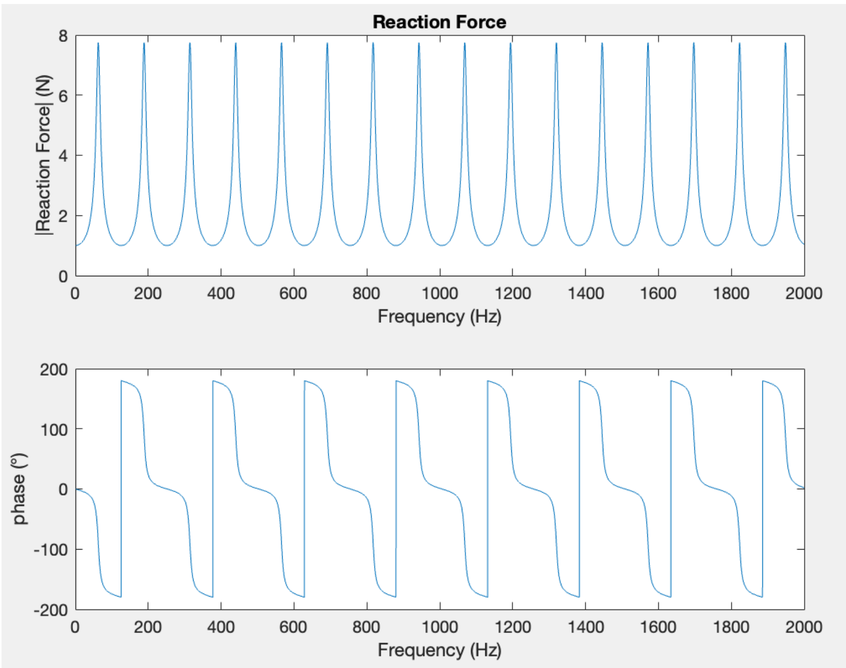


Figure 5: Magnitude and phase of the reaction force $R(\omega)$ transmitted to the plate. Each peak corresponds to a resonance of the string. The phase undergoes typical 180° shifts around each resonance.

From the figure, we can clearly notice that the behavior is the same as the Transfer Receptance, with a multiplying factor k_1, because the reaction force is equal to the elastic force.